

Si Yudong

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▣ Education

Tongji University

2019.09 - 2022.03

Nantong University

2014.09 - 2018.03

▣ Work Experience

AI Framework Engineer (Triton Compiler)@Intel.

2022.07 - 2025.07

Triton Compiler XPU Backend Development

- **Feature Implementation**

- Implemented global scratch memory addressing in backend launcher
- Built SPIR-V → Level Zero → SYCL AOT compilation pipeline, facilitating low-level issue reproduction.
- Created Intel specific Pass tritonintelgpu-rewrite-stack-ptr to optimize shared local memory (SLM) pointer handling.
- Integrated __spirv_RoundFToTF32INTEL SPIRV extension for enhanced FP32 precision in tl.dot DPAS lowering.

- **Performance Optimization**

- Established Triton-XPU's first benchmark system covering Softmax/GEMM/FA key kernels, with extensible support for vendor libraries (XeTLA/CUTLASS/oneDNN).
- Landing optimization passes like: Coalesce, AcceletateMatmul, RemoveLayoutConversion, Pipeline / Prefetch, Swizzling etc.
- Optimized critical kernels (e.g. GEMM) and achieved >90% performance of Intel's XeTLA library

- **Bug Fixes**

- Resolved 35+ High-priority compiler backend issues including: performance regressions, Layout propagation failures in IR passes, ut issues and so on.

PyTorch Ecosystem Optimization

- **CI/CD Innovation**

- Designed AWS Xeon-based Jenkins pipeline for PyTorch Inductor CPU Performance, automated collection of 150+ model performance metrics

▣ Project Experience

Embedded Computer Vision System

- **Driver Fatigue Detection System (Jetson Nano and IMX6UL)**
 - Face detection: Traditional HOG + SVM algorithm
 - Key point localization: A 26-point localization model trained based on cascading regression trees
 - Fatigue state detection: Implementation of PERCLOS algorithm and Euler angle calculation to identify features such as eye closure, yawning, and head tilting
 - Alerting: Development of an operational interface using the QT, integration with in-vehicle LIN ambient lighting for alert functionality

▣ Skills

- **Compiler:** Triton,LLVM,MLIR,SPIRV
- **Heterogeneous Computing:** oneAPI,OpenCL,CUDA,SYCL,Level Zero,ROCm
- **AI Framework:** Pytorch,IPEX
- **AI Infrastructure:** Jenkins,GitHub Actions,Docker,Vtune Profiler
- **Languages:** C/C++, Python, Bash